Kyle **Dahlin**

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Professional Experience

2026-present	**Assistant Professor**, Department of Mathematics, Cal Poly
2023-2026	**NSF-MPS Ascending Postdoctoral Research Fellow**, Department of Mathematics, Virginia Tech
2020-2023	**Postdoctoral Research Associate**, Odum School of Ecology and Center for the Ecology of Infectious Diseases, University of Georgia
2018	**Visiting Scholar**, Pacific Biosciences Research Center, University of Hawai'i at Mānoa
2015-2020	**Graduate Teaching Assistant**, Department of Mathematics, Purdue University

Education

Purdue University *West Lafayette, IN*
Ph.D., Mathematics *2020*

- Dissertation title: Mathematical Models for Mosquito-borne Infectious Diseases of Wildlife
- Advisor: Zhilan Feng

M.S., Mathematics *2015*

University of Hawai'i at Mānoa *Honolulu, HI*
B.A., Mathematics *2013*

Publications

For most of my published work (exceptions 1. and 10.), author order indicates level of contribution, with the highest level of contribution associated with the first and last author.

** = joint first authorship, ♯ = graduate student co-author, ♮ = undergraduate co-author*

Pre-prints

13. **Dahlin, Kyle**, Suzanne M. O'Regan, John Paul Schmidt, Barbara A. Han, and John M. Drake. 2024. "Fast-Lived Vertebrate Hosts Exhibit Higher Potential for Mosquito-Borne Parasite Transmission." ***bioRxiv***. doi.org/10.1101/2024.10.21.619438.

Peer-Reviewed

12. Gibbs♯, Theo L., **Kyle Dahlin**, Joe Brennan♯, Cynthia B. Silveira, and Lisa C. McManus (2024). "Coexistence of Bacteria with a Competition-Colonization Tradeoff on a Dynamic Coral Host." ***Theoretical Ecology*** 19 (1).. doi.org/10.1007/s12080-026-00637-y.

11. **Dahlin, Kyle**, Karin Ebey♯, John E. Vinson, and John M. Drake. 2025. "Nonlinear Effects of Noise on Outbreaks of Mosquito-Borne Diseases." ***PLoS Computational Biology***. 22 (4): e1013466. doi.org/10.1371/journal.pcbi.1013466.

10. Cummings, Joseph, **Kyle J-M Dahlin**, Elizabeth Gross, and Jonathan D. Hauenstein. 2025. "Routing Functions for Parameter Space Decomposition to Describe Stability Landscapes of Ecological Models." ***Bulletin of Mathematical Biology*** 87 (12): 177. doi.org/10.1007/s11538-025-01554-7.

9. **Dahlin, Kyle J-M**, Michael A. Robert, and Lauren M. Childs. 2025. "Once Bitten, Twice Shy: A Modeling Framework for Incorporating Heterogeneous Mosquito Biting into Transmission Models." ***Bulletin of Mathematical Biology*** 87 (163): 1–45. doi.org/10.1007/s11538-025-01540-z.

8. *Ferretti, Pamela, Maria M. Martignoni*, Lisa C. McManus, Taom Sakal, Armun Liaghat♯, Bethany Stevens, **Kyle J-M Dahlin**, et al. 2025. "Theory of Host-Microbe Symbioses: Challenges and Opportunities." ***Cell Host & Microbe*** 33 (7): 1052–56. doi.org/10.1016/j.chom.2025.05.001.

7. Reano, Darryl, Dominique David-Chavez, Kinsey Skillen, Kenneth Ridgway, **Kyle Dahlin**, and Rylan Chong. 2025. "Supporting the next Generation of Indigenous STEM Scholars Demands Indigenous Leadership." ***The Journal of Geography*** 124 (3): 62–72. doi.org/10.1080/00221341.2025.2499822.

6. ***Dahlin, Kyle**, Elisa Van Cleemput*, Subodh Adhikari, Karen Castillioni, and Lynette R. Strickland. 2025. "Simulated Biodiversity Hotspots from Traditional Ecological Knowledge and Western Metrics Do Not Always Overlap." ***Nature Communications Earth & Environment*** 6 (1): 1–9. doi.org/10.1038/s43247-025-02309-x.

5. **Dahlin, Kyle**, Suzanne M. O'Regan, Barbara A. Han, John Paul Schmidt, and John M. Drake. 2024. "Impacts of Host Availability and Temperature on Mosquito-borne Parasite Transmission." ***Ecological Monographs*** 94 (2). doi.org/10.1002/ecm.1603.

4. *,♯Howerton, Emily, **Kyle Dahlin***, Christina J. Edholm, Lindsey Fox, Margaret Reynolds, Brandon Hollingsworth, George Lytle, Melody Walker, Julie Blackwood, and Suzanne Lenhart. 2023. "The Effect of Governance Structures on Optimal Control of Two-Patch Epidemic Models." ***Journal of Mathematical Biology*** 87 (5): 74. doi.org/10.1007/s00285-023-02001-8.

3. Drake, John M., **Kyle Dahlin**, Pejman Rohani, and Andreas Handel. 2021. "Five Approaches to the Suppression of SARS-CoV-2 without Intensive Social Distancing." ***Proceedings of the Royal Society B: Biological Sciences*** 288 (1949): 20203074. https://doi.org/10.1098/rspb.2020.3074.

2. **Dahlin, Kyle**, and Zhilan Feng. 2019. "Modeling the Population Impacts of Avian Malaria on Hawaiian Honeycreepers: Bifurcation Analysis and Implications for Conservation." ***Mathematical Biosciences*** 318 (108268): 108268. doi.org/10.1016/j.mbs.2019.108268.

1. **Dahlin, Kyle**, Erika König, Amanda Laubmeier, Austin Wehn, and Karen Rios-Soto. 2012. "Competition Model between the Invasive Sahara Mustard and Native Plants in the Sonoran Desert". Mathematical and Theoretical Biology Institute Technical Reports 09-01M 2012.

General audience

Dahlin, Kyle, and Stacy M. T. Potes. 2025. "'Āina Momona: The Proliferation of Place-Based Math Pedagogy in Hawai'i." ***MAA Focus*** 45 (2): 34–36.

In prep

3. *Modeling interactions between mosquito biting persistence and defensive behaviors on mosquito-borne disease transmission* with Michael Robert, Lauren Childs, and Clément Vinauger

2. *Optimal control of vectored, venereal, and vertical La Crosse Virus transmission in a wildlife setting* with George Lytle and Benjamin Lebdaoui (undergraduate)

1. *Explainable identification of animal disease reservoirs through topological data analysis* with Sarah Percival

Awards, Fellowships, & Grants

Research Grants

2026	**Arctic Lens Mini-Grant**, Center for Sustainable Engagement in the Arctic, Virginia Tech *(PI, K. Dahlin)*	*$1,000*
2025	**Seed Grant**, Pandemic Prediction and Prevention Destination Area, Virginia Tech *(PI, K. Dahlin, co-PIs Clément Vinauger, Lauren M. Childs, Michael Robert)*	*$20,000*
2023	**Ascending Postdoctoral Fellowship (DMS-2316455)**, NSF Directorate for the Mathematical and Physical Sciences *(PI, K. Dahlin)*	*$300,000*
2021	**Modelling University Responses to COVID-19**, NIH-MIDAS Coordination Center *(co-PIs, K. Dahlin and John Drake)*	*$9,939*

Other Grants

2025	**Career Development Grant**, Sloan Scholars Mentoring Network *(PI, K. Dahlin)*	*$3,000*
	Seminar Speaker Support, VT Center for the Mathematics of Biosystems	*$500*
2023	**Tensor SUMMA Grant**, Mathematical Association of America *to the Indigenous Mathematicians Network*	*$6,000*

Honors & Awards

2022-2024 **MGB-SIAM Early Career Fellowship**, Society for Industrial and Applied Mathematics

2021 **Contributed Talk Award for Mathematical Epidemiology**,
Society for Mathematical Biology & University of California, Riverside

2019 **Excellence in Teaching Award**, Purdue University Department of Mathematics

2018 **Cagiantas Fellowship**, Purdue University
· Awarded to "a senior Ph.D. student who has demonstrated a high level of accomplishment in their research and has participated in activities that have had a positive impact on the climate of their department, University, or community."

2017 **AGEP Mentoring Award**, Purdue University

2016 **Lighting the Pathway Fellowship**, American Indian Science and Engineering Society

2015-2019 **AGEP Scholarship**, Purdue University

2013-2019 **Sloan Indigenous Graduate Partnership Scholarship**, Sloan Foundation

2012 **Undergraduate Student Poster Presentation Award**, SACNAS

2010 **Undergraduate Math and Biology Research Program Fellowship**,
University of Hawai'i at Mānoa

Presentations

Colloquia

2025	Virginia Tech Mathematics Colloquium	*Blacksburg, VA*
	James Madison University Mathematics & Statistics Colloquium	*Harrisonburg, VA*
	Mathematics and Climate Research Network	*Virtual*

Invited Seminar Presentations

2026	University of Montana Sloan Indigenous Graduate Program Seminar	*Virtual*
2025	Virginia Tech Ecology and Evolutionary Biology Seminar	*Blacksburg, VA*
	University of Maryland Mathematical Biology Seminar	*College Park, MA*
	University of New Mexico Applied Math Seminar	*Albuquerque, NM*
	Virginia Commonwealth University BioMath Seminar	*Richmond, VA*
2024	Virginia Tech Mathematical Biology Seminar	*Blacksburg, VA*
2023	University of Hawai'i at Mānoa Applied Math Seminar	*Honolulu, HI*
2022	University of California, Davis Mathematical Biology Seminar	*Virtual*
2021	MIDAS Webinar: COVID-19 Urgent Grant Program Awards	*Virtual*
	Virginia Tech Mathematical Biology Seminar	*Virtual*
2020	Center for the Ecology of Infectious Diseases Disease Ecology Workshop	*Athens, GA*
2018	University of Hawai'i at West O'ahu	*Kapolei, HI*
	University of Wollongong	*Wollongong, Australia*
	Purdue University Student Colloquium	*West Lafayette, IN*
2017	Purdue University Student Colloquium	*West Lafayette, IN*
2016	Purdue University Graduate Student Research Day	*West Lafayette, IN*
	Purdue University Student Colloquium	*West Lafayette, IN*

Invited Conference Presentations

2026	Joint Mathematics Meetings	*Washington, DC*
2025	Society for Mathematical Biology Annual Conference	*Edmonton, Canada*
	Joint Mathematics Meetings	*Seattle, WA*
2024	Society for Mathematical Biology Annual Meeting	*Seoul, South Korea*
	AISES Region 7 Conference	*Hampton, VA*
	Joint Mathematics Meetings	*San Francisco, CA*
2022	SIAM Annual Meeting	*Pittsburgh, PA*
	Joint Mathematics Meetings	*Virtual*
2019	SACNAS National Conference	*Honolulu, HI*
	Society for Mathematical Biology Annual Conference	*Montreal, Canada*
	AMS Spring Central and Western Joint Sectional Meeting	*Honolulu, HI*
2018	6th International Conference on Mathematical Biology	*Beijing, China*
	AMS Spring Central Sectional Meeting	*Columbus, OH*

Contributed Presentations

2025	Infectious Diseases Research Symposium, Virginia Tech Center for Emerging, Zoonotic, and Arthropod-borne Pathogens (CeZAP)	*Blacksburg, VA*
2023	Infectious Diseases Research Symposium, CeZAP	*Blacksburg, VA*
2021	Society for Mathematical Biology Annual Conference	*Virtual*
2020	SACNAS National Conference	*Virtual*
2018	AISES National Conference	*Oklahoma City, OK*
2012	SACNAS National Conference	*Seattle, WA*

Poster Presentations

2023	Virginia Tech Mechanistic Biological Modeling Symposium	*Blacksburg, VA*
2022	Society for Vector Ecology Annual Conference	*Honolulu, HI*
	Ecology & Evolution of Infectious Diseases	*Atlanta, GA*
2018	The Wildlife Society Annual Conference	*Reno, NV*
	Society for Mathematical Biology Annual Conference	*Sydney, Australia*

Symposia Organized

2025	ʻĀina Momona: A symposium on place-based math education in Hawaiʻi, with Stacy Potes	*Honolulu, HI*
2024	Society for Mathematical Biology Annual Meeting: Postdoctoral Research Showcase, with Laura Strube	*Seoul, South Korea*
2023	SIAM Conference on Control and Its Applications: Applications of Control in Biological Systems, with Suzanne Lenhart	*Philadelphia, PA*
2021	Society for Mathematical Biology Annual Meeting: Ecological models at the interface of empirical and theoretical research, with Amanda Laubmeier	*Virtual*
2019	Society for Mathematical Biology Annual Meeting: Mathematical models for infectious diseases at population and individual levels, with Joan Ponce	*Montreal, Canada*
2018	Society for Mathematical Biology Annual Meeting: Mathematical modeling of malaria: Dynamics within-host and between-hosts, with Lauren Childs	*Sydney, Australia*
	6th International Conference on Mathematical Biology: Mathematical Models in Ecology, with Joan Ponce	*Beijing, China*

Public Outreach

2023 University of Hawai'i at Mānoa 808 State Math Challenge — *Ewa Beach, HI*
University of Hawai'i at Mānoa Undergraduate Math Club — *Honolulu, HI*
2018 Hands of the Future, Inc. — *West Lafayette, IN*

Teaching Experience

Cal Poly

Fall 2026 Instructor, **Linear Analysis (MATH 2341)**

Virginia Tech

Spring 2026 Project leader, **Center for the Mathematics of Biosystems Summer School**
Spring 2026 Associate Teaching Badge, **Center for the Integration of Research, Teaching, and Learning**
Fall 2025 Instructor of record, **Introduction to Linear Algebra (MATH 2114)**
Guest lecturer, **Infectious Disease Ecology (BIOL 4564)**
Fall 2024 Guest lecturer, **Mathematical Methods for Modeling Biological Systems (MATH 5515)**

University of Georgia

Summer 2023 Research mentor, **Population biology of infectious diseases (NSF Research Experiences for Undergraduates)**
Summer 2022 Instructor, **Simulating infectious diseases with *R* (workshop)**

Purdue University

Summer 2019 Course coordinator & Instructor, **Calculus I (MA 161)**
Spring 2018 Online instructional support, **Applied Calculus I & II (MA 16010 & 16020)**
Fall 2017 Instructor, **Applied Calculus I (MA 16010)**
Spring 2017 Instructor, **Calculus II, Active Learning (MA 162)**
Fall 2016 Instructor, **Calculus I, Active Learning (MA 161)**
Summer 2016 Instructor, **Calculus I (MA 161)**
Spring 2016 Instructor, **Applied Calculus II (MA 16020)**
Fall 2015 Recitation Instructor, **Multivariable Calculus (MA 261)**
Summer 2015-2019 Instructor, **GRE Quantitative Exam Preparation for AGEP**

University of Hawai'i at Mānoa

Summer 2024 Organizer, **Place-based Mathematics Education Seminar**
2012 - 2013 Calculus and Physics Tutor, **Native Hawaiian Science and Engineering Mentorship Program**
2012 - 2013 Teaching Assistant, **Natural Sciences Undergraduate Learning Emporium**

Mentoring

At Cal Poly

N. Hazzard (undergraduate, Mathematics, Cal Poly): Mentored senior thesis project.

At Virginia Tech

A. Miller (graduate student, Mathematics, Virginia Tech): Met weekly to discuss integral projection modeling and computational techniques for coral population dynamics.

T. Gibbs (graduate student, Quantitative and Computational Biology, Princeton) and J. Brennan (graduate student, Population Biology, UC Davis): Provided guidance on the development, simulation, and analysis of models of population dynamics of coral holobionts.

At University of Georgia

A. Willoughby (graduate student, Ecology): Provided guidance on the development on a model of rodent population cycles.

K. Ebey (undergraduate REU, Biology, Chemistry, and Mathematics): Advised on a project using stochastic differential equations to model the effects of environmental variation on mosquito-borne disease transmission.

N. Tabassum (graduate student, Mathematics, Texas Tech): Provided guidance on generating thermal performance curves for mosquito traits and integrating them into modeling.

At Purdue University

Four undergraduates (LSAMP) and five graduate students (Graduate Bridge Program) in a wide variety of disciplines (Mathematics, Statistics, Physics, Wildlife Biology and Conservation, Electrical and Computer Engineering, Mechanical Engineering, Chemical Engineering, and Mechatronics Engineering) through the AGEP program.

Leadership & Service

Professional Service

2025-2026	Annual Meeting Organizing Committee, **Society for Industrial and Applied Mathematics (SIAM)**
2024-2026	Chair, Workshop Celebrating Diversity Working Group, **SIAM**
2025	Faculty advisor, **American Indian Science and Engineering Society (AISES) at Virginia Tech**
2023-present	Member, **Virginia Tech Indigenous Caucus**
2021-present	Member-at-Large, Membership Committee, **Society for Mathematical Biology**
2020-present	Co-director, **Indigenous Mathematicians Network**
2024-2025	Conference Organizing Committee, **Mid-Atlantic Regional Math Alliance**
2014-2018	Vice President (2014); Treasurer (2015-2018), **AISES, Purdue Chapter**
2016-2017	Graduate Representative, **Purdue University Department of Mathematics**
2013-2016	Assistant Coach, **C. Richard Petticrew Forum (Purdue Collegiate Debate Team)**
2012-2013	Captain, **University of Hawai'i at Mānoa Debate Society**

Professional Memberships

American Mathematical Society (AMS)
Society for Industrial and Applied Mathematics (SIAM)
Society for Mathematical Biology (SMB)
Mathematical Association of America (MAA)
Association for Women in Mathematics (AWM)
Spectra
American Indian Science and Engineering Society (AISES)

Workshops Attended

2025 **+Policy Network Policy Camp**, Virginia Tech
The Math Bundle, Banff International Research Station
New Collaborations Workshop, Center for the Mathematics of Biosystems, Virginia Tech

2024 **Theory of Microbial Symbiosis Workshop**, Hawai'i Institute of Marine Biology, University of Hawai'i at Mānoa

2021 **VectorBiTE 2021 Training**, VectorBiTE Research Coordination Network
Dynamics of Infectious Diseases: Ecological Models Across Multiple Scales, AMS Mathematics Research Community

2020 **Workshop for New Investigators**, NSF Directorate for Mathematical and Physical Sciences. 2020.
Adaptive Management Tutorial, National Institute for Mathematical and Biological Synthesis, University of Tennessee, Knoxville

2018 **Applied Management Principles**, Krannert Executive Education Programs, Purdue University
Disease Ecology and Eco-epidemiology, Mathematical Biosciences Institute, Ohio State University

2017 **Industrial Mathematical and Statistical Modeling Workshop**, Statistical and Applied Mathematical Sciences Institute, North Carolina State University

2016 **Dynamics of Biological Systems**, Seminaire de Mathematiques Superieure, University of Alberta

2015 **Uncertainty, Sensitivity, and Predictability in Ecology: Mathematical Challenges and Ecological Applications**, Mathematical Biosciences Institute, Ohio State University

Journals Refereed

Bulletin of Mathematical Biology · Journal of Mathematical Biology · Mathematical Biosciences · Ecology Letters
Journal of Theoretical Biology · Methods in Ecology and Evolution · Proceedings of the Royal Society B
PLoS Computational Biology

Grant Review Panels

National Science Foundation, Division of Mathematical Sciences (2025)

Languages

Proficient in MATLAB, R, Julia, and LaTeX. Intermediate experience with Python. English (native) and Hawaiian (intermediate).

Other Certifications

Wilderness First Aid (MEDIC SOLO), Exp. 4/27
Basic Life Support (Red Cross), Exp. 6/28